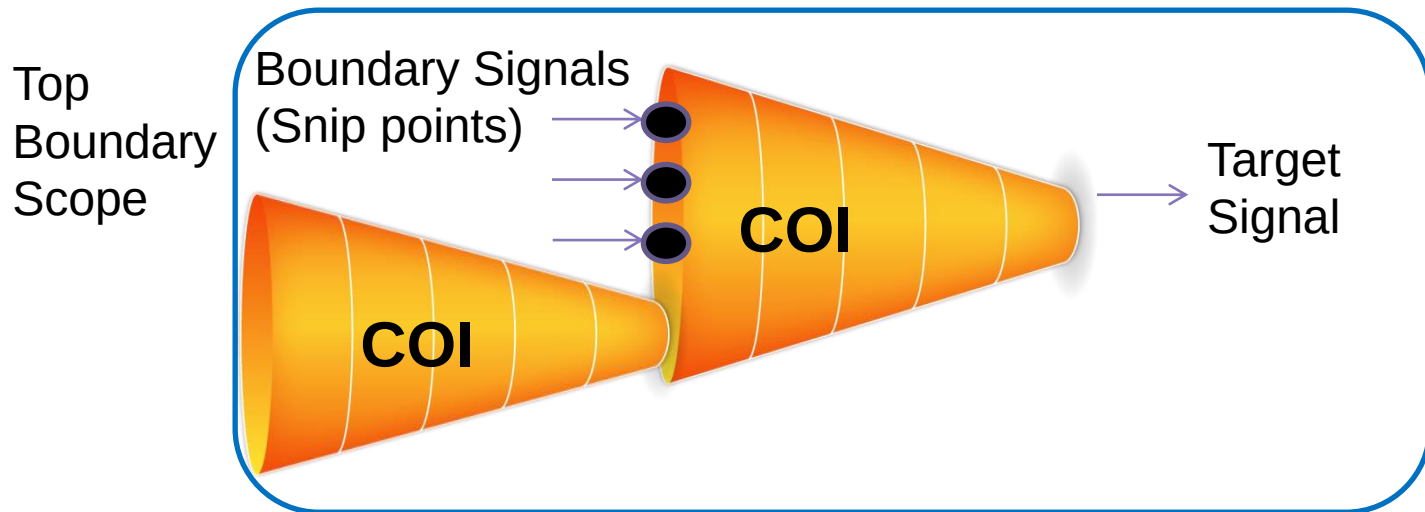


Outline

- Overview
- COI Main Window
- Preference
- Output GUI
- COI batch mode - Input TCL command

COI Marking – Overview

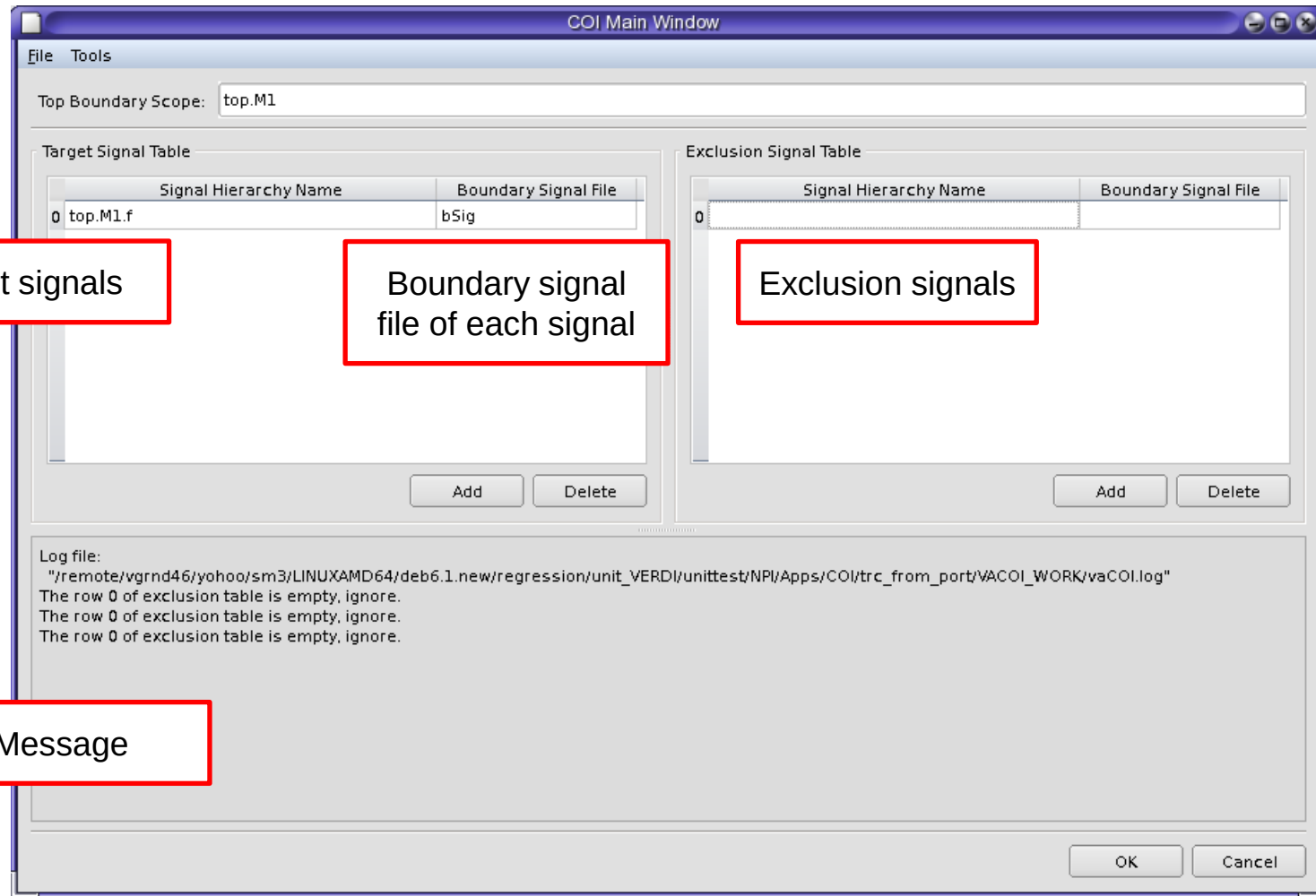


- Target Signal
 - The full hierarchical name of signals that the App will analyze their cone of influence (COI)
- Boundary Signal(s)
 - The signals that the App will stop the tracing of COI
- Top Boundary Scope
 - The top scope that the App will stop the tracing
- Exclusion Signal (optional)
 - Remove the cone from the COI of target signal

Outline

- Overview
- COI Main Window
- Preference
- Output GUI
- COI batch mode - Input TCL command

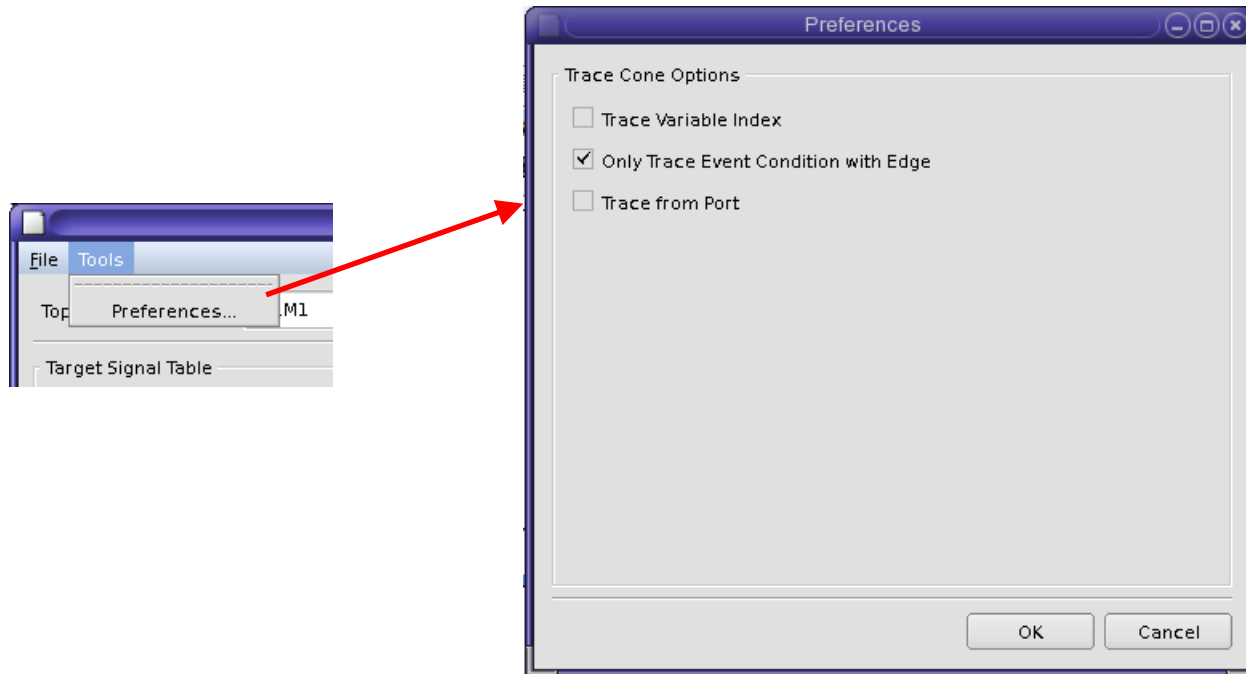
COI Main Window



Outline

- Overview
- COI Main Window
- **Preference**
- Output GUI
- COI batch mode - Input TCL command

Preference



Preference (cont.)

- Trace Variable Index
 - Treat index variable as an driver.
 - For driver signal “g[a]”, both “g” and “a” are drivers.

```
13
14 module nm(inout f,
15           inout l,
16           input m);
17   wire a, b, c, d, e;
18   wire [1:0] g;
19   assign g = b + c;
20   assign e = a;
21   assign f = g[e];
22   assign l = m;
23   nm N1(b, c);
```

Preference (cont.)

- Only Trace Event Condition with Edge
 - Only trace edge trigger signal of event condition.
 - For “always@(aa)”, “aa” is not a driver due to it's not edge trigger signal.

```
4 module mn(output reg h,  
5           output i);  
6  
7 wire aa = 5;  
8 wire bb= 6;  
9 assign i = 0;  
10 always@(aa)  
11     h = bb;  
12 endmodule
```

Preference (cont.)

- Trace from Port
 - Treat each target signal as port.
 - For the tracing from port “top.M1.f”, the line 14 doesn’t be highlighted. But it’ll be highlighted if tracing from signal “top.M1.f”

```
14 module mm(inout f,  
15           inout l,  
16           input m);  
17   wire a, b, c, d, e ;  
18   wire [1:0] g;  
19   assign g = b + c;  
20   assign e = a;  
21   assign f = g[e];  
22   assign l = m;  
23   nn N1(b, c);  
24  
25 endmodule
```

Outline

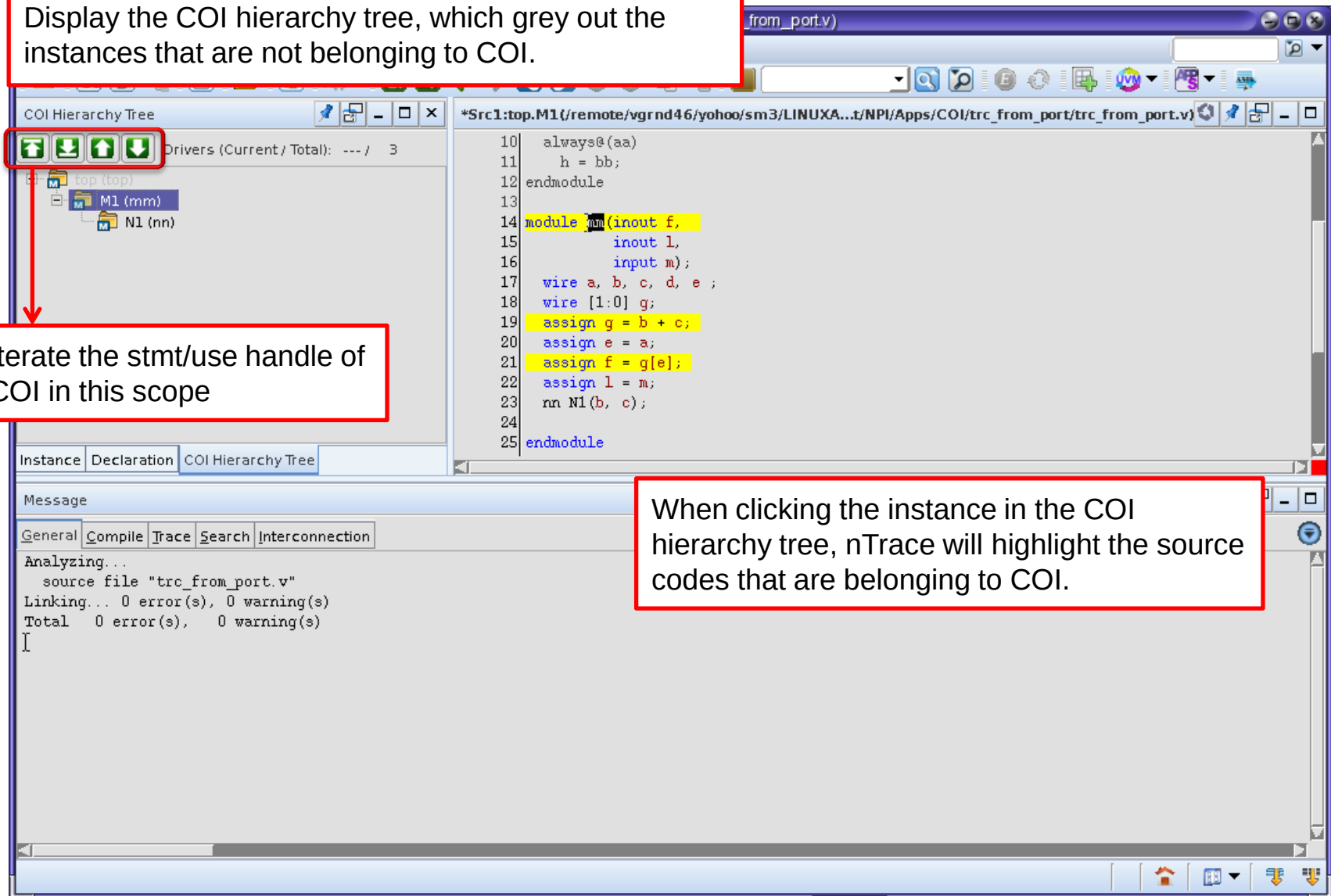
- Overview
- COI Main Window
- Preference
- **Output GUI**
- COI batch mode - Input TCL command

Output GUI

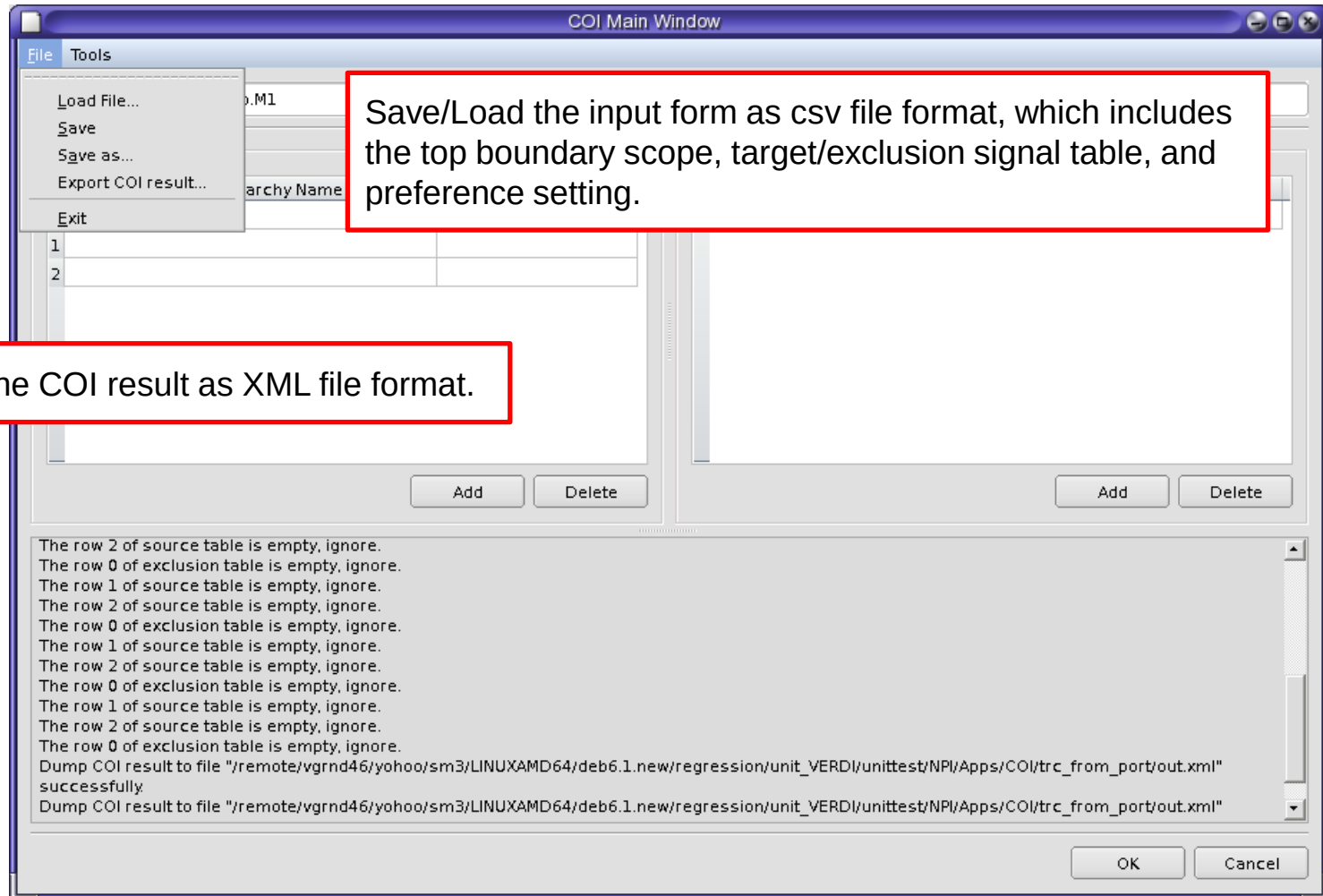
Display the COI hierarchy tree, which grey out the instances that are not belonging to COI.

Iterate the stmt/use handle of COI in this scope

When clicking the instance in the COI hierarchy tree, nTrace will highlight the source codes that are belonging to COI.



Restore/Export File



COI result by XML

```
<COIresult>
<Scope Type="npiModule" FullName="top.M1.N1" Location="/remote/vgrnd46/yohoo/sm3/LINUXAMD64/deb6.1.new/regression/unit_VERDI/unittest/NPI/Apps/COI/trc_from_port/trc_from_port.v : 23">
  <Stmt Type="npiPort" Name="h" Location="/remote/vgrnd46/yohoo/sm3/LINUXAMD64/deb6.1.new/regression/unit_VERDI/unittest/NPI/Apps/COI/trc_from_port/trc_from_port.v : 4"/>
  <Stmt Type="npiPort" Name="i" Location="/remote/vgrnd46/yohoo/sm3/LINUXAMD64/deb6.1.new/regression/unit_VERDI/unittest/NPI/Apps/COI/trc_from_port/trc_from_port.v : 5"/>
  <Stmt Type="npiContAssign" Name="assign bb = 6" Location="/remote/vgrnd46/yohoo/sm3/LINUXAMD64/deb6.1.new/regression/unit_VERDI/unittest/NPI/Apps/COI/trc_from_port/trc_from_port.v : 8"/>
  <Stmt Type="npiContAssign" Name="assign i = 0" Location="/remote/vgrnd46/yohoo/sm3/LINUXAMD64/deb6.1.new/regression/unit_VERDI/unittest/NPI/Apps/COI/trc_from_port/trc_from_port.v : 9"/>
  <Stmt Type="npiEventControl" Name="@aa" Location="/remote/vgrnd46/yohoo/sm3/LINUXAMD64/deb6.1.new/regression/unit_VERDI/unittest/NPI/Apps/COI/trc_from_port/trc_from_port.v : 10"/>
  <Stmt Type="npiAssignment" Name="h = bb;" Location="/remote/vgrnd46/yohoo/sm3/LINUXAMD64/deb6.1.new/regression/unit_VERDI/unittest/NPI/Apps/COI/trc_from_port/trc_from_port.v : 11"/>
</Scope>
<Scope Type="npiModule" FullName="top" Location="/remote/vgrnd46/yohoo/sm3/LINUXAMD64/deb6.1.new/regression/unit_VERDI/unittest/NPI/Apps/COI/trc_from_port/trc_from_port.v : 28">
  <Stmt Type="npiContAssign" Name="assign k = j" Location="/remote/vgrnd46/yohoo/sm3/LINUXAMD64/deb6.1.new/regression/unit_VERDI/unittest/NPI/Apps/COI/trc_from_port/trc_from_port.v : 31"/>
</Scope>
<Scope Type="npiModule" FullName="top.M1" Location="/remote/vgrnd46/yohoo/sm3/LINUXAMD64/deb6.1.new/regression/unit_VERDI/unittest/NPI/Apps/COI/trc_from_port/trc_from_port.v : 34">
  <Stmt Type="npiPort" Name="f" Location="/remote/vgrnd46/yohoo/sm3/LINUXAMD64/deb6.1.new/regression/unit_VERDI/unittest/NPI/Apps/COI/trc_from_port/trc_from_port.v : 14"/>
  <Stmt Type="npiContAssign" Name="assign g = (b + c)" Location="/remote/vgrnd46/yohoo/sm3/LINUXAMD64/deb6.1.new/regression/unit_VERDI/unittest/NPI/Apps/COI/trc_from_port/trc_from_port.v : 19"/>
  <Stmt Type="npiContAssign" Name="assign f = g[e]" Location="/remote/vgrnd46/yohoo/sm3/LINUXAMD64/deb6.1.new/regression/unit_VERDI/unittest/NPI/Apps/COI/trc_from_port/trc_from_port.v : 21"/>
</Scope>
</COIresult>
```

Outline

- Overview
- COI Main Window
- Preference
- Output GUI
- COI batch mode - Input TCL command

Input TCL Format

- `npi_coi_mark { targetSigFile exclusionSigFile
topBoundaryScope outputLog trcOptionList COIresult}`
 - `targetFileName`
 - The file name which describes the target signal(s) and the boundary signals file. Refer the “COI File” format
 - `exclusionFileName` (optional)
 - The file name which describes the target signal(s) and the boundary signals file. Refer the “COI File” format
 - `topBoundaryScope` (optional)
 - The top scope full hier. name that the COI stops at
 - `outputLog` (optional)
 - The output log file name

Input TCL Format

- np_i_coi_mark { targetSigFile exclusionSigFile
topBoundaryScope outputLog trcOptionList COIresult}
 - trcOptionList (optional)
 - Specify the tracing rule of COI, the valid rule keywords could be:
NPI_COI_TRC_VAR_IDX
NPI_COI_NO_EDGE_CHECK
NPI_COI_TRC_FROM_PORT
 - Refer to the Preference.
 - COIresult (optional)
 - Export the COI result as XML file format, Refer the COI result by XML.

COI File – Format

boundary_signals_file
can be empty

<signal_full_hier_name> [<boundary_signals_file_name>]



one signal per line

<signal_full_hier_name>
<signal_full_hier_name>
...

For example:

targetFile

top.f[2] boundary_1.list
top.f[3:4] boundary_2.list
top.g



boundary_1.list

top.m0.mo[3:2]
top.m1.i1

Input Tcl – Only One Target Signal

- If you only have one target signal, no boundary signals, then the content of “targetFileName” can be

target_file.txt

top.inst1.inst4.End

- So the COI marking Tcl can be
 - Tcl> np_i_coi_mark “target_file.txt” “” “Top”